

Unpacking the internal assessment

Criterion B: Planning

Decomposing problems

This part of the computational thinking process is important in breaking down a complex problem into a number of more manageable problems. The decomposition will also inform planning, and any plan will involve the order and requirements of each of the decomposed problems.

Decomposition involves identifying the main components of the solution as well as its functional parts. Once the main components have been clearly identified, students should divide each component into smaller and more specific tasks. This approach reduces complexity, making each task easier to tackle.

Decomposing problems into smaller, manageable sections is a fundamental step in the computational thinking process. By doing this, students can better plan and execute the development of their computational solution.

There are multiple tools to facilitate decomposition.

- The general solution can be documented by a use case diagram together with flow charts, data flow diagrams and/or the equivalent.
- For procedural coding it may be helpful to use a flow-charting tool to break down a complex task into modules.
- For object-oriented coding it is important to identify and describe the objects in the problem scenario.
- A database project requires identifying and describing the data, and the data processing needed.

Planning the computational thinking process

Once a problem has been decomposed it is possible to plan for its solution. The plan should outline the order of tasks, the timeline for their completion and the resources needed to complete individual tasks. Any research that is required can also be planned for.

The plan should clearly address the steps involved in all five stages of:

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- planning
- designing
- developing
- testing
- evaluating.

A GANTT chart can be used to map out the steps in the plan. It is also important to maintain an AGILE approach where possible, so that the documentation matches the reality of the proposed solution and addresses the details. A generic GANTT chart will not fulfil this criterion.

After decomposing a problem, students should create a detailed chronological plan, considering:

- task order and dependencies, to ensure efficient task execution
- success criteria to address all requirements for a successful solution
- subtasks that break down major activities into steps for planning, designing, developing, testing and evaluating
- time, limitations and resources to allocate realistic timeframes, consider constraints and identify resource needs
- planning of research to integrate any necessary research activities within the overall plan.

Students need to be clear that the plan is not a process journal. Also, while an AGILE approach is encouraged, the planning stage of the documentation should not be written after the computational solution is completed, but beforehand—it is a plan from the outset that guides development of the solution.

